OUTLINE AND DRILL MAP scale 4.4 : 1 seen from the FRONT (component side of U1, odd fingers, finger 1 at the left)

Datum: lower-left corner of the card rectangle = origin of the Gerber, drill, netlist and placement files (x right, y up). Tolerances of the DDR4 SO-DIMM module drawing (JEDEC MO-310 as reproduced in references/socket, Foxconn 317-0000-2253 sheet 2).

A CORNER CHAMFER 0.6 x 45 deg (4x)

B LATCH NOTCH (2x, both ends): y 18.00 .. 22.00, 1.75 deep, inner corners R0.65

C TOOLING HOLE (2x): dia 1.80 NON-PLATED at (2.00, 6.00) and (67.60, 6.00)

D KEY SLOT: 1.00 +/-0.05 wide (x 37.80 .. 38.80), 4.00 +/-0.10 deep, full radius R0.50 at its end

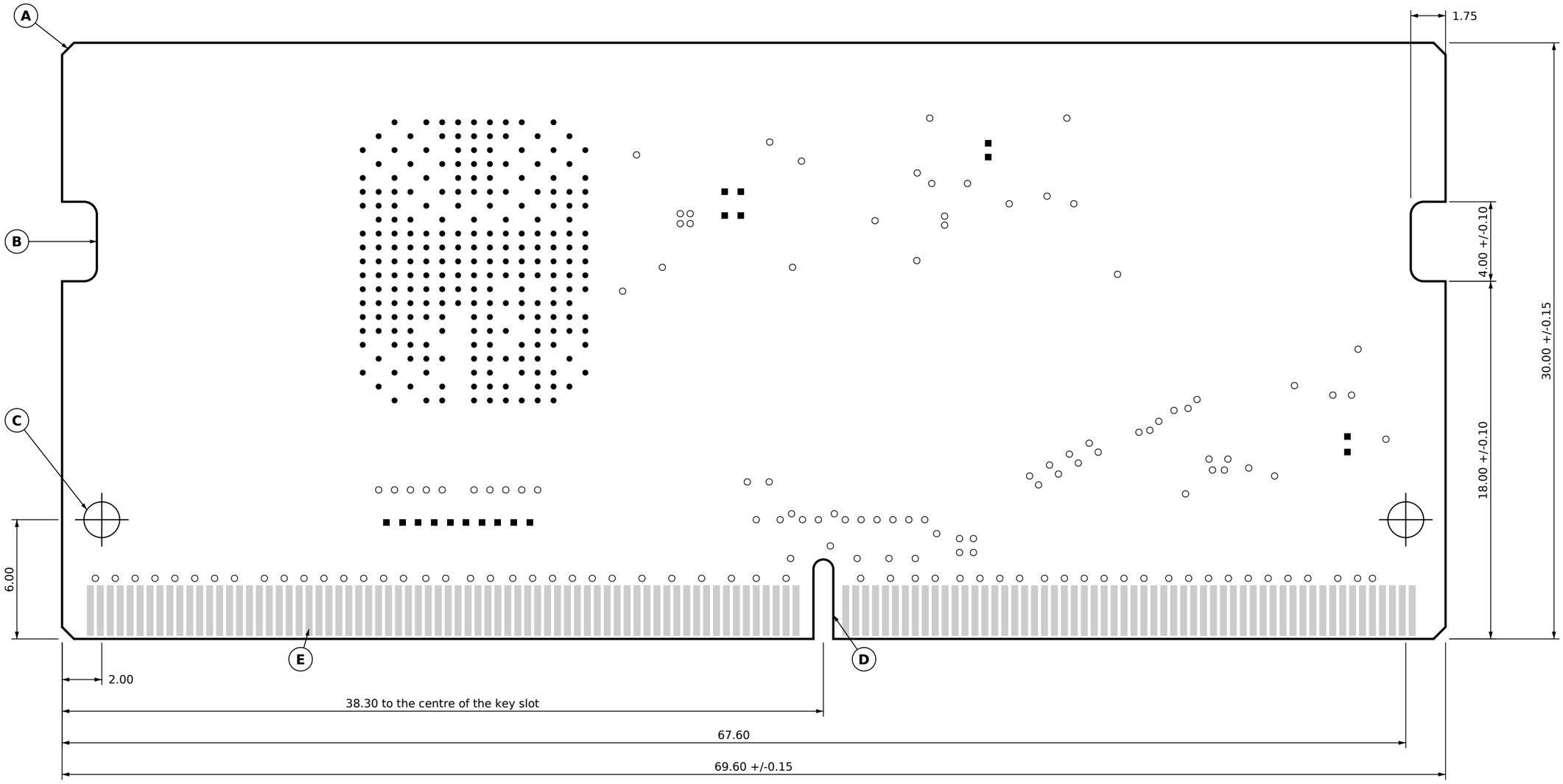
E FINGERS: 260 lands, 130 per face, 0.35 +/-0.03 wide, pitch 0.50 per face, y = 0.15 .. 2.70 above the edge; finger 1 (front) centre x = 1.425, finger 2 (back) x = 1.675: the back-side row is offset 0.25 mm

DETAIL: FINGER EDGE (section, not to scale)



FINISHED THICKNESS measured OVER THE GOLD FINGERS: absolute tolerance, not +/-10 %

Chamfer 0.25 x 45 deg on both faces, no deep bevel. Hard gold over nickel on the lands, which start <= 0.30 mm from the edge. No solder mask on the finger strip (y < 2.75 mm).



DRILL TABLE (finished hole sizes; files in fab/drill/)

Symbol	Dia mm	Plating	Qty	Pad mm	Use / treatment
solid dot	0.15	plated	235	0.40	via-in-pad of U1: FILLED AND CAPPED (IPC-4761 type VII), flush
solid square	0.20	plated	18	0.40 / 0.45	FILLED AND CAPPED: 10 command node vias (y = 5.85), 8 vias in the exposed pads of U2 / U4 / U5
open circle	0.20	plated	140	0.40 / 0.45	plain via, tented on both faces
crossed circle	1.80	NON-plated	2	-	tooling holes of the card edge

Plated holes 393 (235 x 0.15 mm, 158 x 0.20 mm), non-plated 2. Filled and capped 253: listed one by one in fab/drill/filled_capped_vias.csv and lpddr5-testbed-filled_capped.drl.
Through holes only: no blind, buried or micro vias. Smallest annular ring 0.10 mm (0.40 mm pad on 0.20 mm, 0.35 mm In3 pad on 0.15 mm at U1 balls B3 / C4).

LPDDR5-CMD-TESTBED ZCU104 R0.2

FABRICATION DRAWING: outline, drill

lpddr5-testbed.kicad_pcb sheet 1 of 3 dimensions in mm do not scale

board file SHA-256 ad7fe7fb7fc2977633b94e019d2710f9...

STACKUP (6 layers, top to bottom)

Layer	Material	mm	er	Finished / copper density	Role
F.Mask	LPI solder mask	0.010	3.5	0.020 over copper	black (green allowed, note 12); removed over the fingers
L1 F.Cu	copper	0.035	-	finished 0.040 nominal with plating; 16 % copper	signal + parts (microstrip over L2), no pour
	prepreg PR2116	0.110	4.29		impedance-defining for F.Cu (unchanged)
L2 In1.Cu	copper	0.018	-	92 % copper	GND plane
	core FR4	0.360	4.29		thinned from 0.400 to 0.360 for the 1.20 mm card
L3 In2.Cu	copper	0.018	-	94 % copper	GND plane + 1.2 V island under the VDD fingers
	prepreg PR2116	0.110	4.29		impedance-defining for In3 (unchanged)
L4 In3.Cu	copper	0.018	-	2 % copper	high-speed stripline only
	core FR4	0.360	4.29		thinned from 0.400 to 0.360 for the 1.20 mm card
L5 In4.Cu	copper	0.018	-	92 % copper	GND plane + VDD2H island under U1
	prepreg PR2116	0.110	4.29		impedance-defining for B.Cu (unchanged)
L6 B.Cu	copper	0.035	-	finished 0.040 nominal with plating; 16 % copper	signal + parts (microstrip over L5), power pours under U1
B.Mask	LPI solder mask	0.010	3.5	0.020 over copper	black (green allowed, note 12); removed over the fingers

Total 1.212 mm with mask, 1.192 mm at the bare fingers, 1.20 mm with hard gold. REQUIRED: finished thickness measured over the gold fingers 1.20 +/-0.10 mm.

The two FR4 cores are 0.36 mm (14 mil); the three PR2116 prepregs (0.11 mm) set the impedances and must not be changed. With cap plating of the filled vias the outer copper may reach 0.055 mm: 1.24 mm.

CONTROLLED IMPEDANCE (all +/-10 %; the fab may adjust the widths by +/-0.02 mm; TDR coupons on the panel border, without copper flood beside the lines)

Layer	Reference planes	Type	Width mm	Gap mm	Design ohm	Order ohm	Nets
In3.Cu	L3 / L5 (stripline)	single	0.15	-	45.6	46	CA0-6, CS0; read DQ0-7
In3.Cu	L3 / L5 (stripline)	differential	0.10	0.10	84.6	85	CK_P / CK_N; RDQS_P / RDQS_N
F.Cu, B.Cu	L2 / L5 (microstrip)	single	0.20	-	45.6	46	finger stubs, read finger tracks
F.Cu	L2 (microstrip)	single	0.15	-	52.4	52	DQ0-7 lines from U1 to the dividers
F.Cu, B.Cu	L2 / L5 (microstrip)	differential	0.10	0.10	91.8	92	WCK pair (B.Cu); last 5.8 mm of RDQS (F.Cu)

Outer-layer design values are for 0.040 mm finished copper under 0.020 mm of mask, er 4.29 (2-D field solver, design/impedance_results.json).

VIA TREATMENT

FILLED AND CAPPED, IPC-4761 type VII, cap flush within +/-0.025 mm: 253 vias = 235 x 0.40 / 0.15 mm in the lands of U1 + 18 x 0.20 mm drill listed below.

ALL OTHER VIAS (140) are plain and tented on both faces. The 0.20 mm filled holes share their drill size with plain vias: they are identified by position.

UNUSED INNER-LAYER PADS ARE REMOVED IN THE DATA (do not add them back) on every via except the ten command node vias (row y = 5.85 mm).

Hole to copper 0.20 mm is used at its limit beside the RDQS pair inside U1 (0.205 mm) and at its two return vias above fingers 219 / 221 (0.225 mm).

x mm	y mm	Drill	Pad	Net	Where
16.325	5.850	0.20	0.40	CA5	command divider node via (0201 lands 0.05-0.10 mm from its pad)
17.125	5.850	0.20	0.40	CA6	command divider node via (0201 lands 0.05-0.10 mm from its pad)
17.925	5.850	0.20	0.40	CA3	command divider node via (0201 lands 0.05-0.10 mm from its pad)
18.725	5.850	0.20	0.40	CA4	command divider node via (0201 lands 0.05-0.10 mm from its pad)
19.525	5.850	0.20	0.40	CK_N	command divider node via (0201 lands 0.05-0.10 mm from its pad)
20.325	5.850	0.20	0.40	CK_P	command divider node via (0201 lands 0.05-0.10 mm from its pad)
21.125	5.850	0.20	0.40	CA2	command divider node via (0201 lands 0.05-0.10 mm from its pad)
21.925	5.850	0.20	0.40	CS0	command divider node via (0201 lands 0.05-0.10 mm from its pad)
22.725	5.850	0.20	0.40	CA1	command divider node via (0201 lands 0.05-0.10 mm from its pad)

x mm	y mm	Drill	Pad	Net	Where
23.525	5.850	0.20	0.40	CA0	command divider node via (0201 lands 0.05-0.10 mm from its pad)
33.350	21.300	0.20	0.45	GND	in the land of U2.11
33.350	22.500	0.20	0.45	GND	in the land of U2.11
34.150	21.300	0.20	0.45	GND	in the land of U2.11
34.150	22.500	0.20	0.45	GND	in the land of U2.11
46.600	24.250	0.20	0.45	GND	in the land of U4.7
46.600	24.950	0.20	0.45	GND	in the land of U4.7
64.650	9.400	0.20	0.45	GND	in the land of U5.9
64.650	10.200	0.20	0.45	GND	in the land of U5.9

LPDDR5-CMD-TESTBED ZCU104 R0.2

FABRICATION DRAWING: stackup, impedance, vias

lpddr5-testbed.kicad_pcb sheet 2 of 3 dimensions in mm do not scale

board file SHA-256 ad7fe7fb7fc2977633b94e019d2710f9...

FABRICATION NOTES (the text of layer User.Comments of the board file, verbatim; also in fab/gerber/lpddr5-testbed-User_Comments.gbr)

1. 6-layer FR-4 (Tg >= 170 C). FINISHED THICKNESS MEASURED OVER THE GOLD FINGERS 1.20 +/-0.10 mm (not +/-10 %).

2. STACKUP, top to bottom (mm):

F.Mask	solder mask 0.010 (0.020 over copper), removed over the fingers	
F.Cu	copper 0.035 (finished, with plating: 0.040 nominal)	
	prepreg PR2116 0.110, er 4.29	
In1.Cu	copper 0.018	
	core FR4 0.360, er 4.29	
In2.Cu	copper 0.018	
	prepreg PR2116 0.110, er 4.29	
In3.Cu	copper 0.018	
	core FR4 0.360, er 4.29	
In4.Cu	copper 0.018	
	prepreg PR2116 0.110, er 4.29	
B.Cu	copper 0.035 (finished, with plating: 0.040 nominal)	
B.Mask	solder mask 0.010 (0.020 over copper), removed over the fingers	
total 1.212 with mask, 1.192 at the bare fingers, 1.20 with hard gold.		

3. CONTROLLED IMPEDANCE, all +/-10 %; widths may be adjusted by +/-0.02 mm:

In3.Cu	0.15 mm single (ref In2 / In4)	46 ohm
In3.Cu	0.10 / 0.10 mm pair	85 ohm differential
F.Cu / B.Cu	0.20 mm single (ref In1 / In4)	46 ohm
F.Cu / B.Cu	0.15 mm single	52 ohm
F.Cu / B.Cu	0.10 / 0.10 mm pair (WCK on B.Cu, RDQS drop on F.Cu)	92 ohm differential

4. FINISH: fingers (J1 pads 1 .. 260) hard gold over nickel (30 uin min); all other exposed copper ENIG.

5. CARD EDGE: 0.25 mm x 45 deg chamfer only (no deep bevel); finger copper starts <= 0.30 mm from the edge. No solder mask and no copper pour over the fingers; no plating tie bars left on the card.

6. VIA-IN-PAD, FILLED AND CAPPED (IPC-4761 type VII), 253 vias: the 235 vias 0.40 / 0.15 mm in the lands of U1, the 10 command node vias 0.40 / 0.20 mm (row y = 5.85 mm, x = 16.325 + 0.8 k mm, k = 0 .. 9, from the pin-1 corner) and the 8 vias 0.45 / 0.20 mm in the exposed pads of U2, U4, U5 at (x, y) mm: (33.35, 21.30) (33.35, 22.50) (34.15, 21.30) (34.15, 22.50) (46.60, 24.25) (46.60, 24.95) (64.65, 9.40) (64.65, 10.20).
   The 0.20 mm filled holes share their drill size with plain vias: they are the ones listed here (the board file carries the filled / capped attribute per via).
   All other vias are plain and tented on both sides.

7. UNUSED INNER-LAYER PADS ARE REMOVED IN THE DATA on the via-in-pads of U1, the command fence vias, the node vias of the data side (row y = 6.00 mm), the shunt, read, return and reference vias, the finger vias and every via of the power stage: do not add them back. The ten command node vias (row y = 5.85 mm) keep their pads on every layer. Hole to copper 0.20 mm is used at its limit beside the RDQS pair inside U1 and at its two return vias above fingers 219 / 221 (0.225 mm).

8. Minimum track 0.09 mm (inside U1 only, 0.10 mm elsewhere), minimum gap 0.10 mm, minimum drill 0.15 mm (via-in-pad), 0.20 mm elsewhere; annular ring 0.10 mm.

9. OUTER-LAYER COPPER DENSITY IS LOW BY DESIGN: F.Cu and B.Cu carry tracks, pads and a few small power pours under U1 (about 16 to 17 % of the card area per face, a third of it the finger row); there is NO ground flood. DO NOT ADD COPPER THIEVING, DUMMY FILL OR ANY POUR inside the card outline: the outer-layer lines are impedance-controlled against In1 / In4 without coplanar copper. Balance the plating with robber bars on the panel border; the inner layers In1 / In2 / In4 are solid planes (more than 90 % copper), In3 is a sparse signal layer between them.

10. ACCEPTANCE: IPC-6012 class 2 (annular ring 0.10 mm on the 0.20 mm drills: class 3 is not attainable). 100 % bare-board netlist test against the IPC-D-356 netlist. Bow and twist <= 0.5 %, flatness over the finger row <= 0.15 mm. Cap plating of the filled vias flush within +/-0.025 mm.
    TDR coupons for the impedance classes of note 3 on the panel border, with outer-layer copper density like the card (no flood).

11. PANEL: the card is 30 mm high and must ship in a panel. Rout the outline; break-off tabs on the top edge or on the sides above the latch notches only (y > 22 mm), none on the finger edge or in the notches; no V-score. Panel rails carry three fiducials per face.

12. SOLDER MASK: LPI, 0.05 mm expansion in the data; colour as specified (black) or green if the 0.08 mm dam between the two lands of an 0201 cannot be held in black; where a dam cannot be held, the two lands of one part may share a window. The mask-free finger strip ends at y = 2.75 mm; the row of vias 0.15 mm above the finger tops (y = 3.05 mm, tented, 0.10 mm dam) is intentional: do not move or remove them.

13. ASSEMBLY: three fiducials per face on the card (FID1 .. FID6: 1.0 mm dot, 2.0 mm mask opening). 0.10 mm stencil; no paste on the fingers, test pads and fiducials. Tallest part 1.10 mm on either face (module limit 1.20 mm per face).

LPDDR5-CMD-TESTBED ZCU104 R0.2

FABRICATION DRAWING: notes

lpddr5-testbed.kicad_pcb sheet 3 of 3 dimensions in mm do not scale

board file SHA-256 ad7fe7fb7fc2977633b94e019d2710f9...